

PATIENT DIGITAL TWIN · INTEGRATED NEUROLOGICAL DECISION SUPPORT

Patient Digital Twin Integrated Clinical Report

Multimodal synthesis of clinical, laboratory, imaging, electrophysiology and pharmacogenomic data, cross-referenced against external biomedical evidence per the NeuroTwin Multimodal Biomedical Resource Guide (document date 8 Sep 2026).

Patient & Clinical Context			9/9 data categories included
PATIENT John Doe ID: SYN-100482	AGE / SEX 34 years · Male	PRIMARY DIAGNOSIS Focal impaired-awareness seizures	DIAGNOSED SINCE 12 May 2021
SECONDARY DIAGNOSIS Secondary generalized tonic-clonic seizures			
CURRENT THERAPY Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID			
VITAL SIGNS Not reported No vital-sign dataset (BP, HR, temperature) is present in the source data for this patient.			
REPORTING PERIOD Clinic visits: 01 Jan – 02 Mar 2026 · Labs/imaging/EEG/ECG/questionnaire: 01 Jan – 30 Jun 2026 · PGx panel: reported 18 Aug 2026			

5 Clinical Visits	12 CBC Reports	6 CT Studies	35 EEG Recordings
12 ECG Sessions	6 MRI Series	76 PGx Variants	12 Questionnaires

How to read this report: Every finding below states **Patient Evidence** (observed, from this patient's own records), **External Evidence** (published/database context, cited with source and retrieval basis), **Interpretation** (what the two together plausibly mean), and **Clinical Action** (what a clinician might do next) — followed by a GREEN / AMBER / RED rating with the specific reason for that rating. Database associations, predictions and pathway links are never treated as a definitive patient-specific conclusion; where a resource was not queried, this is stated rather than approximated.

02 Longitudinal Clinical Timeline

DATE	VISIT STATUS	SUMMARY
2026-01-01	stable outpatient follow-up	7 total reported episodes; average duration 3.4 minutes; clustering no. Clinician impression: seizure control reasonably stable, quality-of-life trend
2026-01-16	seizure clustering documented	4 total reported episodes; average duration 9.0 minutes; clustering yes. Clinician impression: seizure control reasonably stable, quality-of-life tren
2026-02-03	stable outpatient follow-up	5 total reported episodes; average duration 6.7 minutes; clustering no. Clinician impression: seizure control reasonably stable, quality-of-life trend
2026-02-15	highest recorded seizure burden	11 total reported episodes; average duration 8.2 minutes; clustering no. Clinician impression: seizure control reasonably stable, quality-of-life tren
2026-03-02	stable outpatient follow-up	5 total reported episodes; average duration 4.5 minutes; clustering no. Clinician impression: seizure control reasonably stable, quality-of-life trend
18 Aug 2026	Pharmacogenomic panel reported	76 variants analyzed across 47 medications (genetics_clinical_summary.json).

03 Diagnoses & Clinical Findings

Diagnosis & Disease Behavior

Observed

PATTERN

Chronic focal epilepsy with focal impaired-awareness seizures and secondary generalized tonic-clonic seizures.

CONTROL

partial control with ongoing breakthrough seizures

NEURO. DECLINE

No

FUNCTIONAL DECLINE

No

Seizure Analysis

Observed

HIGHEST BURDEN

11 episodes, 2026-02-15

LOWEST BURDEN

4 episodes, 2026-01-16

CLUSTERING

Yes (documented once)

Aura / Prodromal Patterns

Patient-reported

Deja vu frequency

Epigastric rising sensation

Feeling of derealization/detachment

Metallic taste in mouth

Racing heartbeat prior to episode

Sudden dizziness before seizure onset

Sudden intense anxiety or fear

Tingling or numbness sensations

Unusual smells (olfactory aura)

Visual distortions or flashing lights

04

Medication & Treatment Profile

DRUG	DOSE
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

Effectiveness: partial response · **Tolerability:** generally tolerated with intermittent dizziness and variable side-effect burden · **Documented toxicity:** No

Persistent Reported Side Effects

Patient-reported

Memory fog post-dose

Tremor in hands

Weight change since last visit

Gum hypertrophy

Mood irritability

Nausea after taking medication

Skin rash or itching

Daytime drowsiness

Dizziness/lightheadedness

Unsteady gait (ataxia)

Integrated Findings

Reported skin rash/itching vs. lamotrigine boxed-warning rash risk

RED

RED because a patient-reported symptom overlaps directly with an FDA boxed-warning event category for a drug the patient is actively taking.

PATIENT EVIDENCE	Clinical notes list "Skin rash or itching" among persistent reported side effects across the 5-visit window (clinical_notes_summary.json → persistent_side_effects). No dermatology assessment, rash description, or discontinuation is recorded in the available data.
EXTERNAL EVIDENCE	[WEB-VERIFIED] FDA prescribing information for lamotrigine carries a Boxed Warning: life-threatening serious rashes, including Stevens-Johnson syndrome and toxic epidermal necrolysis, and rash-related death have occurred; nearly all cases begin within 2–8 weeks of starting therapy; label directs discontinuation at first sign of rash unless clearly unrelated.
INTERPRETATION	A reported rash on lamotrigine cannot be assumed benign from this dataset alone. The boxed warning does not establish that this patient's rash is SJS/TEN — it establishes that lamotrigine-associated rash requires urgent, specific clinical characterization, which is not present in the available records.
CLINICAL ACTION	Recommended: Treating clinician should urgently characterize the reported rash (morphology, mucosal involvement, fever, timing relative to dose changes) and follow the lamotrigine label's rash-management guidance before the next dose.

Source / confidence: FDA-approved lamotrigine labeling (accessdata.fda.gov, Lamictal label PDF; 2025 revision), retrieved via web search this session. Evidence type: regulatory boxed warning. Confidence: high (regulatory-grade evidence); applicability to this specific patient's rash: unresolved (rash character not recorded).

CNS adverse-effect cluster (dizziness, ataxia, tremor, somnolence) vs. known drug profiles

AMBER

AMBER because the symptom cluster is a well-documented, expected class effect of both drugs rather than an unexpected or high-severity signal.

PATIENT EVIDENCE	Persistent side effects include dizziness/lightheadedness, unsteady gait (ataxia), tremor in hands, daytime drowsiness and memory fog; tolerability recorded as "generally tolerated with intermittent dizziness and variable side-effect burden."
EXTERNAL EVIDENCE	[WEB-VERIFIED] DailyMed/FDA labeling lists dizziness and somnolence among the most common adverse reactions for levetiracetam; dizziness, ataxia and somnolence are listed among the most common, dose-related adverse reactions for lamotrigine.
INTERPRETATION	The reported symptom pattern is consistent with expected, dose-related CNS effects of this dual-AED regimen rather than a novel or unexplained toxicity.
CLINICAL ACTION	Continue routine tolerability monitoring; consider dose-timing adjustment if symptoms affect function. No urgent action indicated on this finding alone.

Source / confidence: DailyMed levetiracetam label (setid a05bba4-c34f-4134-8207-db79e4c70445) and lamotrigine/Lamictal labels (dailymed.nlm.nih.gov), retrieved via web search this session. Evidence type: regulatory labeling, class-level. Confidence: high for the class effect; not specific to this patient's exact symptom severity.

05

Laboratory / Vital Trends

CBC (Observed): 12 CBC report(s) from 2026-01-01 to 2026-06-30 were processed. No CBC parameter abnormality was identified against the report-provided reference ranges. Overall status: **STABLE WITHIN REFERENCE RANGES**. No abnormal flags recorded on any of the 12 report dates (2026-01-01 to 2026-06-30).

Vital signs (BP, heart rate, temperature, SpO2): Not available — no vital-sign dataset exists in the source records for this patient.

Stable haematology across 6 months vs. drug-specific haematologic safety profile

GREEN

GREEN because 12/12 consecutive, independently-dated CBC panels show no abnormal flag, and neither current drug carries a haematologic-toxicity boxed warning.

PATIENT EVIDENCE	12 CBC reports over 6 months, zero abnormal flags on any date (CBC_consolidated_summary.json → digital_twin_state.active_flags_by_date).
EXTERNAL EVIDENCE	[WEB-VERIFIED] DailyMed labeling for levetiracetam and lamotrigine lists behavioral/psychiatric effects, hypersensitivity/DRESS and (for lamotrigine) serious rash as their principal boxed/serious-warning categories; neither label centers on a haematologic-toxicity boxed warning of the kind seen with e.g. carbamazepine or valproate.
INTERPRETATION	From a haematologic-safety standpoint, the observed lab trend is consistent with continued tolerance of the current regimen at the current doses.
CLINICAL ACTION	Continue scheduled CBC surveillance at existing cadence; no change indicated by this finding alone.

Source / confidence: Patient data: CBC_consolidated_summary.json (12/12 reports). External: DailyMed labels, retrieved this session. Confidence: high for the negative (no haematologic signal) finding; general labeling review, not a systematic haematotoxicity database query (e.g. no Tox21/CompTox lookup performed).

06

ECG / EEG Findings

ECG

Observed

12 recordings, 99h 11m total, single channel, 256 Hz.

This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

EEG

Observed

35 recordings (12 interictal / 11 preictal / 12 ictal). Digital-twin state: **Chronic Active**, risk level **ELEVATED**, future seizure probability **ELEVATED**.

Elevated EEG-derived seizure risk converges with observed clinical seizure burden

RED

RED because two independent modalities (EEG signal classification and clinician-documented seizure counts) agree, which is a stronger basis for concern than either alone.

PATIENT EVIDENCE	EEG digital-twin risk level and future-seizure-probability both classified Elevated (EEG_clinical_summary.json). Independently, clinical notes document a peak of 11 breakthrough episodes with one clustering event across a 5-visit window on the current dual-AED regimen.
EXTERNAL EVIDENCE	Not applicable / not required — this is a within-patient, cross-modality convergence finding rather than an external-database question. (Per the resource guide's own selection principle: direct patient evidence is the primary basis when two modalities already agree.)
INTERPRETATION	Convergent signal loss-of-seizure-control is more clinically reliable than either the EEG classification or the clinical count alone.
CLINICAL ACTION	Prioritize seizure-frequency and clustering monitoring; consider whether current dosing, adherence, or an adjunct agent should be reassessed at the next visit.

Source / confidence: Patient data only: EEG_clinical_summary.json + clinical_notes_summary.json. Confidence: high (two independent, patient-specific data sources agree). Note: EEG classification is a signal-derived digital-twin state, not a clinician-verified diagnosis (see EEG report's own scope statement).

07MRI / CT & Imaging Findings

CT — 6 studies

Observed

DATE	IMPRESSION
2026-01-12	No acute intracranial abnormality is evident on this single-image model-assisted review.
2026-02-16	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
2026-03-11	No acute intracranial abnormality is evident on this single-image model-assisted review.
2026-04-20	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
2026-05-14	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
2026-06-29	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

Observed

6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

Important limitation:

No radiologist-verified report, lesion segmentation, or hippocampal volumetry is available for either modality. Absence of a visible structural correlate for the patient's breakthrough seizures should be read as *not assessed*, not as a normal scan — consistent with the resource guide's instruction that missing coverage remain unresolved rather than reassuring.

08Pharmacogenomic & Genetic Risk Profile

76 variants across 47 medications · reported 18 Aug 2026

DRUG	GENE (GENOTYPE)	PREDICTED EFFECT	SIGNIFICANCE	CITATION
Lamotrigine	SCN1A (CT)	efficacy	Significant	PMID:28753467
Levetiracetam	SCN1A (CT)	efficacy	Significant	PMID:28753467
Lamotrigine	ABCB1 (GG)	reduced efficacy	Not significant	PMID:32483200
Lamotrigine	CYP3A4 (TT)	reduced efficacy	Significant	PMID:28343093
Lamotrigine	CACNA1H (AG)	efficacy	Significant	PMCID:PMC6171340

Patient's own metabolizer profile also documents **UGT1A4 *1/*3A (decreased function)** and **CYP3A4 (TT)** — the panel links UGT1A4 only to Olanzapine and CYP3A4 to Lamotrigine (see finding below). 10 priority safety flags exist in the panel but none name Levetiracetam or Lamotrigine directly; they apply to other drugs (e.g. Citalopram, Chlorpromazine) not in this patient's current regimen and are therefore **not applicable to current therapy**.

Lamotrigine flagged via CYP3A4 genotype, but published metabolism is UGT-dominant — and the patient's own decreased-function UGT1A4 finding is not linked to lamotrigine in this panel

AMBER

AMBER: a genuine, patient-specific evidentiary gap. The source panel's own mechanistic basis (CYP3A4) is weaker than published pharmacology for this drug, while the patient's own more mechanistically relevant genotype (UGT1A4, decreased function) is not connected to lamotrigine anywhere in the panel.

PATIENT EVIDENCE	Genetics panel: Lamotrigine flagged CYP3A4 (TT) → "reduced efficacy", significant (PMID:28343093). Same panel separately reports this patient's UGT1A4 status as *1/*3A (decreased function), but links that finding only to Olanzapine, not to Lamotrigine.
EXTERNAL EVIDENCE	[WEB-VERIFIED] Published pharmacokinetic literature: lamotrigine is eliminated predominantly by glucuronidation via UGT1A4 (and to a lesser extent UGT2B7/UGT1A3); it is not considered a major CYP450 substrate. CYP3A4 inducers can indirectly increase lamotrigine clearance via UGT induction, but this is a secondary, not primary, mechanism.
INTERPRETATION	The clinical basis for the panel's CYP3A4-linked "reduced efficacy" call for lamotrigine is less well established than a UGT1A4-based mechanism would be. This patient's own decreased-function UGT1A4 genotype is mechanistically more relevant to lamotrigine clearance than the CYP3A4 finding the panel actually cites, but the panel does not draw that connection. This is an interpretation, not a validated conclusion.
CLINICAL ACTION	Recommend clinical pharmacist or genetics review of whether the patient's UGT1A4 *1/*3A genotype has bearing on current lamotrigine clearance/dosing, alongside (not instead of) the panel's own CYP3A4 finding. No dose change is recommended from this finding alone.

Source / confidence: Patient data: genetics_clinical_summary.json (findings_by_therapeutic_class, metabolizer_profile). External: Milosheska et al., Br J Clin Pharmacol 2016 (bpspubs.onlinelibrary.wiley.com/doi/10.1111/bcp.12984) on UGT1A4/UGT2B7 polymorphism and lamotrigine PK, retrieved via web search this session. Evidence type: peer-reviewed PK study + patient genotype. Confidence: moderate — mechanistic plausibility, not a validated CPIC/PharmGKB dosing guideline (none exists for lamotrigine; see Section 09).

SCN1A (CT) called "favorable efficacy" for both current drugs — biologically plausible target-gene link, not a validated actionable marker

AMBER

AMBER: real, mechanistically coherent target-gene relevance exists, but there is no CPIC or equivalent actionable prescribing guideline behind this specific genotype-to-efficacy claim for either drug.

PATIENT EVIDENCE	Genetics panel: SCN1A (CT) → "efficacy" for both Levetiracetam and Lamotrigine, both marked significant (PMID:28753467).
EXTERNAL EVIDENCE	[WEB-VERIFIED] SCN1A encodes the Nav1.1 voltage-gated sodium channel alpha subunit, the single most frequent target of mutations in genetic epilepsy syndromes (NCBI GeneReviews, SCN1A Seizure Disorders); lamotrigine's own mechanism of action is voltage-gated sodium channel blockade, so a target-level rationale exists. No CPIC guideline links SCN1A genotype to levetiracetam or lamotrigine dosing (CPIC's antiepileptic guideline coverage is CYP2C9/HLA-B for phenytoin).
INTERPRETATION	There is a plausible pathway-level rationale (same gene family as the drug's own target), but this does not establish that this specific genotype predicts this patient's actual efficacy — which the patient's own clinical course (partial response, persistent breakthrough seizures) does not clearly support.
CLINICAL ACTION	Treat as supporting mechanistic context only; do not use this finding alone to justify continuing, escalating or changing therapy. Weight the patient's own seizure diary and EEG trend more heavily (Section 06).

Source / confidence: Patient data: genetics_clinical_summary.json. External: NCBI GeneReviews "SCN1A Seizure Disorders" (ncbi.nlm.nih.gov/books/NBK1318) and CPIC guideline index (cpicpgx.org), retrieved via web search this session. Confidence: low-to-moderate for patient-specific efficacy prediction; the gene-target relevance itself is well established, the efficacy-prediction claim is not CPIC/PharmGKB-validated.

09 Drug-Drug Interaction Assessment

NeuroTwin internal DDI screen

Current regimen (Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID) screened against a curated CYP450 reference table (Flockhart-style) and pharmacodynamic class rules already built into this system. Status: **NO INTERACTION DETECTED** Severity: **MINOR**

No resolved pharmacokinetic or pharmacodynamic interaction mechanism was identified between Lamotrigine and Levetiracetam (Keppra) in the available curated and rule-based sources. Predicted-model coverage (SuperCYPsPred) remains unresolved; absence of data does not establish absence of risk.

No curated pharmacokinetic mechanism between Levetiracetam and Lamotrigine

GREEN

GREEN for this specific pairwise mechanism only: neither drug is a significant shared-pathway CYP450 substrate/inhibitor/inducer per curated reference pharmacology, and no pharmacodynamic class rule fired for this pair.

PATIENT EVIDENCE	NeuroTwin's own DDI module (reference_data.py, Flockhart-style curated table) found no shared CYP mechanism between the two current drugs.
EXTERNAL EVIDENCE	[WEB-VERIFIED] Levetiracetam undergoes minimal hepatic/CYP metabolism (predominantly renal elimination); lamotrigine is predominantly UGT-glucuronidated, not CYP-mediated (see Section 08). Independent published pharmacology is consistent with the internal screen's "no shared CYP mechanism" finding.
INTERPRETATION	The absence of a curated pharmacokinetic interaction is corroborated by independent published pharmacology, not just an internal rule with no external cross-check.
CLINICAL ACTION	No DDI-driven dose adjustment indicated between these two agents. A live, comprehensive predictive DDI screen (see below) is still recommended before adding any third agent.

Source / confidence: Patient data: DDI_Clinical_Assessment.json. External: pharmacology literature consistent with above (Section 08 citations). Confidence: moderate-high for this pair; explicitly does not cover future/candidate medications.

Unresolved, not reassuring: SuperCYPsPred predictive-interaction coverage was not evaluated locally for this regimen (no local model snapshot exists in this system). This is reported as *unresolved*, per the resource guide's own instruction that missing coverage must not be interpreted as reassuring.

10 Drug Target & Pathway Analysis

Levetiracetam — SV2A

[WEB-VERIFIED] Binds the synaptic vesicle glycoprotein SV2A, modulating calcium-dependent neurotransmitter exocytosis; a mechanism distinct from other antiseizure drug classes.

Evidence-supported

Source: Lynch et al., PNAS 2004 (pnas.org/doi/10.1073/pnas.0308208101), retrieved via web search this session. Confidence: high (foundational pharmacology paper); general drug-class mechanism, not patient-specific.

Lamotrigine — Voltage-Gated Sodium Channels

[WEB-VERIFIED] Blocks voltage-gated sodium channels, stabilizing neuronal membranes and reducing repetitive firing. Evidence-supported

Source: FDA-approved product labeling (accessdata.fda.gov Lamictal label), retrieved this session. Confidence: high (regulatory-grade mechanism-of-action statement).

Target-gene convergence (Interpretation, not proof of response): Lamotrigine's own target class (voltage-gated Na⁺ channels) overlaps with the gene the panel flagged for both drugs, SCN1A (Nav1.1 alpha subunit). This is a coherent pathway-level rationale for why SCN1A genotype could plausibly modulate lamotrigine response — it is not evidence that it does so in this patient (see Section 08 AMBER finding).

11 Brain / Tissue Biological Relevance

ABCB1 (P-glycoprotein) as a biologically plausible, unconfirmed contributor to breakthrough seizures

AMBER

AMBER: a real, published mechanism exists for how this gene family could contribute to the patient's persistent seizures, but the patient's own panel marks this specific finding "not significant", and no direct causal test was performed.

PATIENT EVIDENCE	Genetics panel: Lamotrigine flagged ABCB1 (GG) → "reduced efficacy", explicitly marked <i>not significant</i> in the source panel. Patient also has persistent breakthrough seizures on dual therapy (Sections 03, 06).
EXTERNAL EVIDENCE	[WEB-VERIFIED] ABCB1 encodes P-glycoprotein, the major ATP-dependent drug-efflux transporter at the blood-brain barrier; most antiepileptic drugs are P-gp substrates, and ABCB1 overexpression/polymorphism has been proposed as a mechanism of pharmacoresistant epilepsy (Loscher & Potschka; multiple case-control studies).
INTERPRETATION	A biologically plausible pathway links ABCB1 genotype to drug-resistant epilepsy in general, but the patient's own panel already rates this specific finding as not significant, and brain-region-specific expression or transporter activity was not measured in this patient. This remains a hypothesis worth noting, not an explanation.
CLINICAL ACTION	No specific action indicated from this finding alone; consider mentioning as background context if breakthrough seizures persist despite therapeutic adherence and adequate dosing.

Source / confidence: Patient data: genetics_clinical_summary.json (ABCB1 finding, marked not significant). External: Sisodiya/Loscher-type ABCB1-pharmacoresistance literature (PMC3125047, pmc.ncbi.nlm.nih.gov), retrieved this session. Confidence: low for patient-specific applicability; the general mechanism is well published, brain-region expression in this patient was not assessed (Human Protein Atlas / GTEx not queried).

Brain-region-specific expression atlases (Human Protein Atlas Brain Atlas, GTEx, Allen Brain Atlas) were **not queried in this session** — these require dedicated gene-level API access beyond a web search and are reported here as not retrieved rather than approximated.

Decision-support synthesis only -- not a substitute for clinician review. Synthetic research data.

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12 Integrated Risk & Benefit Findings

DOMAIN	BENEFIT / GREEN	RISK / RED	UNRESOLVED / AMBER
Seizure control	No progressive decline documented; lowest burden 4 episodes	Peak 11 episodes; 1 clustering event; EEG risk Elevated (convergent, Section 06)	—
Tolerability	No documented drug toxicity; effects mostly known class effects	Reported rash overlaps lamotrigine SJS/TEN boxed warning (Section 04)	Rash character not recorded
Haematologic safety	12/12 CBC reports normal (Section 05)	—	—
Cardiac safety	—	—	No diagnostic rhythm/QT read available (Section 06)
Structural imaging	No acute finding on 6 CT studies	—	No radiologist-verified MRI/CT; no seizure-focus correlate assessed (Section 07)
Pharmacogenomics	SCN1A target-gene plausibility for both drugs	—	CYP3A4 vs UGT1A4 mechanistic gap for lamotrigine; no CPIC guideline exists (Section 08)
Drug–drug interaction	No curated CYP mechanism; corroborated externally (Section 09)	—	Predictive (SuperCYPsPred) coverage unresolved

13 Clinical Action Summary

PRIORITY	ACTION	BASIS
HIGH	Urgently characterize the reported skin rash/itching (morphology, mucosal involvement, timing, fever)	Lamotrigine SJS/TEN boxed warning (Section 04)
HIGH	Reassess seizure control given peak 11-episode burden, one clustering event and Elevated EEG risk	Convergent clinical + EEG evidence (Section 06)
MODERATE	Genetics/pharmacist review of UGT1A4 decreased-function genotype relative to lamotrigine clearance	Section 08 mechanistic gap
MODERATE	Obtain formal radiology review of MRI volumes; consider hippocampal volumetry	No structural seizure-focus correlate assessed (Section 07)
MODERATE	Screen regimen against a live, comprehensive predictive DDI model before any new agent	SuperCYPsPred unresolved (Section 09)
ROUTINE	Continue CBC surveillance at current cadence	12/12 normal to date (Section 05)
ROUTINE	Continue seizure/aura diary and sleep-schedule consistency	Ongoing monitoring priority

14 6-Month Patient Summary

Across the 01 Jan – 30 Jun 2026 observation window, John Doe (34, male) continued dual antiseizure therapy with Levetiracetam 1000 mg BID and Lamotrigine 150 mg BID for chronic focal epilepsy with secondary generalization. Clinician impressions across five visits described reasonably stable seizure control, but breakthrough episodes persisted (range 4–11 per reporting interval) with one documented clustering event, and EEG signal classification independently rated future seizure risk as Elevated. Haematology remained normal across all 12 CBC panels. Ten persistent, mostly dose-related side effects were reported, including a rash/itching finding not further characterized in the available records. Structural imaging (6 CT, 6 MRI series) showed no acute abnormality but was not radiologist-verified. A 76-variant pharmacogenomic panel named both current drugs on SCN1A and flagged lamotrigine additionally via CYP3A4, though published pharmacology attributes lamotrigine clearance mainly to UGT1A4/UGT2B7 — a pathway in which this patient's own panel separately documents a decreased-function UGT1A4 genotype not linked to lamotrigine in the source table. No curated pharmacokinetic interaction was found between the two current drugs.

15 Overall Digital Twin Assessment

Response: Partial. Clinician-documented “partial response” and persistent breakthrough seizures, corroborated by an independently Elevated EEG risk classification, indicate incomplete seizure control on the current regimen — this is the dominant, patient-evidence-driven finding and outweighs the favorable-efficacy pharmacogenomic associations for both drugs (Section 08), which remain hypothesis-level and not CPIC-validated.

Toxicity: No documented drug toxicity and normal haematology support continued tolerability from a laboratory-safety standpoint, but the reported, uncharacterized rash against lamotrigine's boxed SJS/TEN warning is an unresolved, high-priority safety question that external evidence made visible and that must be closed before this domain can be called green.

Functional outcome: No documented neurological or functional decline; cognitive and sleep questionnaire domains trended stable-to-improving over the window, though the lowest-scoring domain (epileptic triggers, 56.7/100) and persistent CNS side effects indicate ongoing quality-of-life burden.

Final interpretation (per resource-guide principle): External biological and regulatory evidence in this report is broadly *consistent with* what the patient's own longitudinal data already shows — it explains plausible mechanisms (SV2A/sodium-channel targets, UGT-vs-CYP clearance, boxed-warning rash risk, P-gp-related pharmacoresistance) but does not itself establish efficacy, safety, or causation for this individual. The patient's own seizure diary, EEG trend, CBC trend and reported symptoms remain the *primary* determinants of this assessment; the two AMBER pharmacogenomic findings and the one RED dermatologic finding are the items requiring clinician follow-up before the next scheduled review.